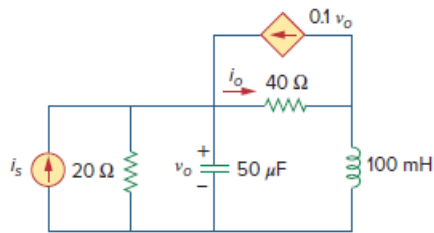


Problem

Use nodal analysis to find current i_o in the circuit of Fig. 10.57. Let $i_s = 6 \cos(200t + 15^\circ)$ A.

Figure 10.57:



Step-by-step solution

Step 1 of 6

$$\omega = 200$$

$$6 \cos(200t + 15^\circ) \Rightarrow 6 \angle 15^\circ$$

$$100 \text{ mH} \Rightarrow j\omega L$$

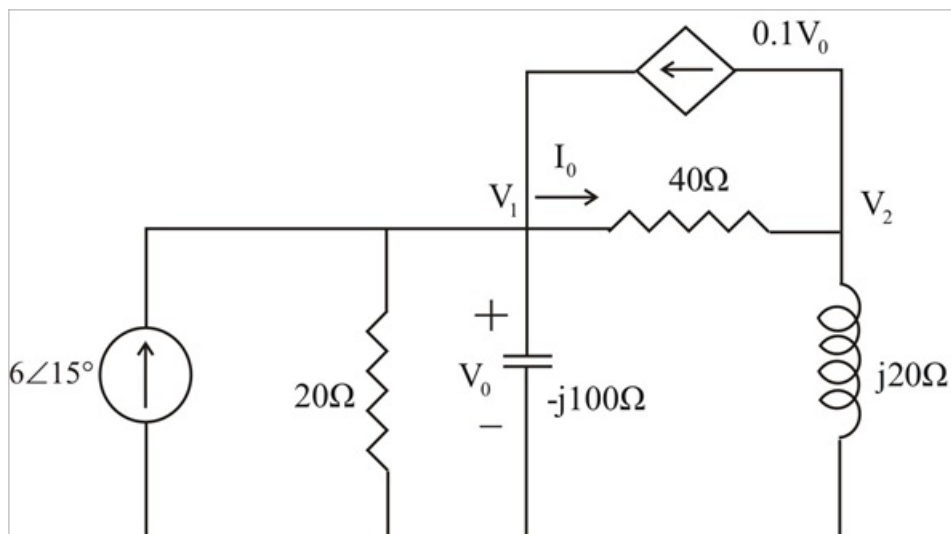
$$j200 \times 0.1 = j20$$

$$50 \mu\text{F} \Rightarrow \frac{1}{j\omega C}$$

$$\frac{1}{j200 \times 50 \times 10^{-6}} = -j100$$

Step 2 of 6

The frequency domain version of the circuit is shown below.



Step 3 of 6

At node 1,

$$6\angle 15^\circ + 0.1V_0 = \frac{V_1}{20} + \frac{V_1}{-j100} + \frac{V_1 - V_2}{40} \text{ Since, } V_0 = V_1$$

$$6\angle 15^\circ + 0.1V_1 = \frac{V_1}{20} + \frac{jV_1}{100} + \frac{V_1 - V_2}{40}$$

$$5.796 + j1.553 = V_1 \left[\frac{1}{20} + \frac{j}{100} + \frac{1}{40} - 0.1 \right] - \frac{V_2}{40}$$

$$5.796 + j1.553 = V_1 [0.05 + 0.01j + 0.025 - 0.1] - \frac{V_2}{40}$$

$$5.796 + j1.553 = (-0.025 + j0.01)V_1 - 0.025V_2 \dots\dots\dots (1)$$

At node 2,

$$\frac{V_1 - V_2}{40} = 0.1V_1 + \frac{V_2}{j20}$$

$$V_1 - V_2 = 4V_1 - 2jV_2$$

$$0 = 3V_1 + (1 - j2)V_2 \dots\dots\dots (2)$$

Step 4 of 6

From Equation-(2),

$$V_2 = \frac{-3V_1}{1 - j2}$$

V_2 is substituted in Equation-(1) we get,

$$5.796 + j1.553 = (-0.025 + j0.01)V_1 - 0.025 \left(\frac{-3V_1}{1 - j2} \right)$$

$$5.796 + j1.553 = V_1 \left[(-0.025 + j0.01) - 0.025 \left(\frac{-3}{1 - j2} \right) \right]$$

$$5.796 + j1.553 = V_1 \left[(-0.025 + j0.01) + \left(\frac{0.075}{1 - j2} \right) \right]$$

$$5.796 + j1.553 = V_1 \left[\frac{-0.005 + j0.06 + 0.075}{1 - j2} \right]$$

$$5.796 + j1.553 = V_1 \left[\frac{0.07 + j0.06}{1 - j2} \right]$$

$$\left(\frac{8.902 - j10.04}{0.07 + j0.06} \right) = V_1$$

$$V_1 = 2.44 - j145.52 \text{ V}$$

Step 5 of 6

From Equation-(2),

$$V_2 = \frac{-3(2.44 - j145.52)}{1 - j2}$$

$$V_2 = \frac{(-7.32 + j436.56)}{1 - j2}$$

$$V_2 = -176.09 + j84.38 \text{ V}$$

Step 6 of 6

Therefore,

$$I_0 = \frac{V_1 - V_2}{40}$$

$$I_0 = \frac{(2.44 - j145.52) - (-176.09 + j84.38)}{40}$$

$$I_0 = \frac{178.53 - j229.9}{40}$$

$$\therefore I_0 = 4.463 - j5.7475 \text{ A} \text{ or}$$

$$\therefore I_0 = 7.276 \angle -52.17^\circ \text{ A}$$